2025 Inventory Strategy & Demand Planning Analysis

Data-Driven Optimization for Supply Chain Resilience

Key Tools: Tableau & Excel-Based Analytics

Chloe Lee

Project Overview & Objectives

Context:

In the fast-paced fashion & beauty industry, balancing stock levels with volatile demand is critical.

Goal:

To analyze 2025 sales trends and lead times to minimize stockouts and optimize inventory turnover.

Key Deliverables:

Demand forecasting, risk assessment, and replenishment strategy.

Demand Trend Analysis

(Insights from Tableau)

Key Visual:

'Monthly Sales & Demand Trend' chart.

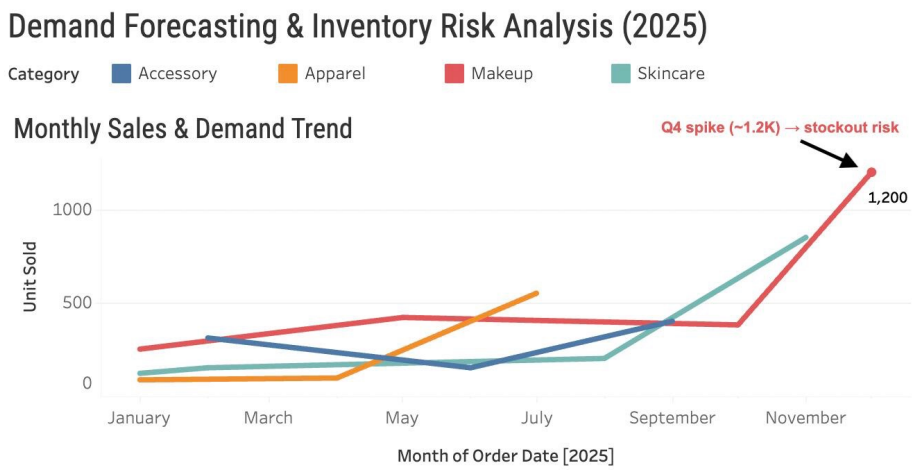
Observations:

* Significant surge in the Makeup category during Q4, peaking at ~1,200 units.

This trend is likely driven by holiday seasonal demand.

Takeaway:

Procurement must be prioritized 2-3 months ahead of Q4 to meet this peak.



Supply Chain & Lead Time Complexity

Key Visual:

Map showing Vendor Locations (South Korea, China, Vietnam, Italy, France).

Deep Dive:

* Analyzing Lead Times is crucial:

While South Korea takes ~15 days, Vietnam and Italy require 28-33 days.

Strategic Insight:

”I didn't just look at stock counts; I integrated lead time data to determine the actual Reorder Point for each SKU."

ROP = (Average Daily Usage X Lead Time) + Safety Stock

Inventory Risk Assessment
(Problem Solving)

Key Insight: Demand Shift

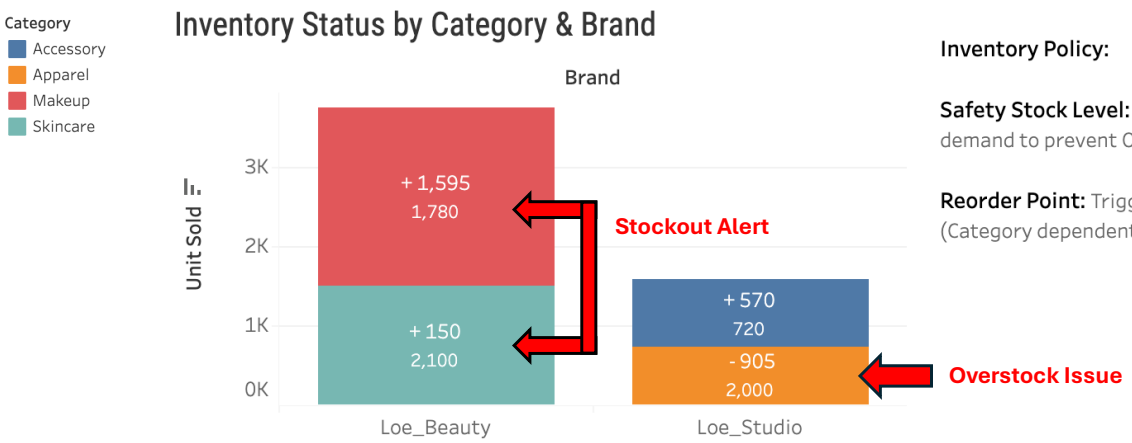
- Observed a significant surge in **Makeup** sales in Q4 (peaking at **~1,200 units**), indicating a sharp holiday demand shift.

Critical Risk 1: Stockout Alert

- Projected shortage in **Skincare & Makeup**, risking lost sales during peak demand.
- Immediate risk items: Velvet Tint (45 units left), Silk Scarf (20 units left).

Critical Risk 2: Overstock Issue

- **Apparel** category holds **905 units** of excess inventory, significantly increasing holding costs.



Inventory Policy:

Safety Stock Level: Maintained at **20%** of average monthly demand to prevent OOS during lead time.

Reorder Point: Triggered when inventory drops below **300 units** (Category dependent).

Revenue Recovery & Inventory Optimization: Strategic Action Plan

Target Impact

- Reduce potential OOS (Out of Stock) risk during peak demand shifts.
- Minimize holding costs by accelerating Apparel liquidation.

Recommendation 1:
Immediate replenishment of +1,770 units for Skincare & Makeup to prevent lost sales.

Recommendation 2:
Implement a 20% Safety Stock policy to buffer against lead time variability.

Recommendation 3:
Launch targeted promotions for the Apparel category to accelerate sell-through.